

Duolingo Tracker

Automated App Feature & User Satisfaction Intelligence Dashboard

ROLE	Data Analyst / Product Analytics	DOMAIN	EdTech / Product Analytics
TOOLS	Python, Pandas, Excel, NLP, SQL Server, Power BI, DAX	DATE	September 2026
AUTHORED BY	Link: https://app.powerbi.com/links/rBx5uoXC8J?ctid=4517da72-c8f7-4cec-b2fc-fda9fe4354f9&pbi_source=linkShare	DATASET	10,000 Google Play reviews and Kaggle dataset

EXECUTIVE SUMMARY	Duolingo Tracker is a product analytics project built to understand how users respond to different features of the Duolingo mobile app. Using 10,000 publicly available Google Play reviews, the project measures overall satisfaction, surfaces the features users discuss most, identifies which topics correlate with lower ratings, tracks how review activity shifts over time, and examines review signals around a specific product event — the March 2025 Dark Mode rollout — as a focused case study. The analysis was automated using Python, structured in SQL Server, and presented through an interactive Power BI dashboard — turning raw review text into structured, decision-ready product insight.
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INTRODUCTION	App reviews contain a large volume of unstructured user feedback, but reading thousands of reviews manually makes it hard to spot recurring product issues. A product team needs answers to questions such as: which features are users talking about the most, which topics receive the lowest ratings, what share of reviews mentioning a feature are negative, and whether feedback shifted around a notable product update. This project builds an end-to-end pipeline — from raw review text to an interactive dashboard — to answer exactly those questions for the Duolingo app.
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STATEMENT OF THE PROBLEM	The core problem: unstructured, high-volume user feedback is difficult to turn into structured, decision-ready product insight without an automated pipeline. Key questions the project set out to answer • Which features are users talking about the most? • Which topics receive the lowest ratings? • What percentage of reviews mentioning a feature are negative? • Did user feedback shift around a notable product update?
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DECISION CRITERIA AND ASSUMPTIONS

The project was scoped around four objectives that guided every downstream decision:

- Measure user satisfaction — track ratings and the positive/negative review split.
- Identify feature-level signals — automatically classify reviews into product topics (Lessons, Pricing, Ads, Hearts/Lives, Streaks, Bugs, Subscription, AI, Dark Mode, and others).
- Detect changes over time — analyze weekly rating trends and shifts in review patterns.
- Investigate product-event signals — examine whether review activity changed around the March 2025 Dark Mode rollout.

Dataset and assumptions

The dataset covers August 19, 2024 – April 9, 2025 (roughly 7.7 months of sampled reviews, not a full year), with fields including reviewId, content, score, thumbsUpCount, reviewCreatedVersion, at, and appVersion. A shift in the sampling pattern was identified around January 13, 2025, so the full period was not treated as one continuous trend — the post-January window was analyzed separately for weekly time-series work.

DATA ANALYSIS

Data preparation (Python)

Python and Pandas were used to clean the dataset, convert dates and timestamps, remove unused fields, handle missing values, derive rating-based sentiment, generate NLP sentiment scores, and automatically classify product topics into analysis-ready datasets.

Sentiment

Star ratings served as the primary sentiment classification (1–2 = Negative, 3 = Neutral, 4–5 = Positive). NLP sentiment via VADER was kept as a separate signal, since text sentiment didn't always agree with the star rating.

Topic classification

A rule-based classification system using regular expressions and keyword patterns sorted reviews into topics such as Pricing, Ads, Hearts/Lives, Practice, Lessons, Streak, Bugs, Subscription, AI, Dark Mode, XP, and Gems. A review can belong to multiple topics, so topic percentages are not mutually exclusive.

SQL data layer

Cleaned data was loaded into Microsoft SQL Server across four tables — reviews (cleaned review-level data), review_topics (normalized review-to-topic mapping), versions (app-version and release-date proxy data), and product_events (documented product-event context) — with relationships and validation checks to ensure topic records mapped correctly back to source reviews.

Power BI dashboard

- Executive Overview — total reviews, average rating, positive/negative %, weekly average rating, rating and sentiment distribution, most discussed topics.
- Feature Intelligence — topic mention volume, negative review share by topic, average rating by topic, topic-level summary matrix.
- Update Impact — Dark Mode as a focused product-event case study (before / event window / after).
- Product Action Center — topic performance summary, volume vs. average rating, high-negative-share topics, dynamic insight cards.

Key findings — overall feedback

Across the sample: 10,000 reviews, 3.28 average rating, 51.2% positive, 34.7% negative — a mixed feedback landscape rather than uniformly positive or negative sentiment. Lessons was the most

frequently mentioned topic (~2,495 reviews), followed by Pricing (2,009), Ads (1,886), Hearts/Lives (1,820), Practice (1,482), and Streak (1,002). Since a review can mention multiple topics, these are topic mentions, not unique review counts.

Topic	Avg. Rating
Dark Mode	1.81
Subscription	2.21
Hearts/Lives	2.30
Pricing	2.32
XP	2.37
Bugs	2.38

These are descriptive associations within the sample — not proof that the feature itself caused the rating.

Dark Mode case study

The clearest event-related pattern in the dataset appeared around March 6–10, 2025, when Dark Mode mentions rose sharply: Mar 6 — 3, Mar 7 — 14, Mar 8 — 11, Mar 9 — 7, Mar 10 — 15, Mar 11 — 4, Mar 12 — 2, Mar 13 — 3.

Period	Avg. Rating	Negative %
Before (Feb 26 – Mar 5)	3.52	29.28%
Event window (Mar 6 – 13)	3.31	33.02%
After (Mar 14 – 20)	3.46	30.76%

The strongest signal is the sharp rise in Dark Mode-related mentions. The rating movement occurred in the same window, but this analysis does not establish a causal relationship between the product event and the rating change.

**ALTERNATIVE
APPROACHES
CONSIDERED**

- Several modelling and classification choices were evaluated before settling on the final approach:
- Substring keyword matching vs. regex word-boundary matching — substring rules created false positives (e.g., “ads” and “AI” misclassified in unrelated contexts); word-boundary regex with narrower keyword lists fixed this.
 - NLP sentiment (VADER) alone vs. star-rating-based sentiment — VADER misclassified some low-star reviews, so star rating was kept as the primary sentiment signal, with VADER retained as a secondary, separate signal rather than the source of truth.
 - Treating the full sample as one continuous time series vs. splitting at the sampling-pattern shift — a shift was detected around January 13, 2025, so the post-January window was analyzed separately for weekly trends instead of smoothing over the discontinuity.

RECOMMENDATIONS

The dashboard is designed to walk a viewer through a clear analytical path: overall satisfaction → feature-level problems → time-based changes → product-event investigation. Within that scope, the analysis is strongest where it stays descriptive and defensible — for example, flagging Dark Mode, Subscription, and Pricing as the lowest-rated topics, and

surfacing the mention spike around the Dark Mode rollout as a pattern worth investigating further.

The analysis intentionally does not claim to have improved retention, identified the exact cause of any rating decline, or prescribed which feature the company should fix — the underlying data doesn't support causal claims of that kind, and the case study is stronger for stopping at what it can actually show.

**IMPLEMENTATION
PLAN (WHAT WAS
BUILT)**

End-to-end pipeline:

Raw Google Play Reviews → Python Data Cleaning → Sentiment + Rule-Based Topic Classification → Analysis-Ready CSV Data → SQL Server Database → SQL Analysis & Validation → Power BI Data Model → DAX Measures → Interactive Product Analytics Dashboard

Skills demonstrated

- Data Analytics: exploratory data analysis, product analytics, KPI design, trend analysis, feature-level analysis
- Python: Pandas, data cleaning, regex, NLP sentiment, automated classification, ETL preparation
- SQL: Microsoft SQL Server, database design, joins, aggregation, CTE/query analysis, data validation, relationships and indexing
- Power BI: data modeling, relationships, DAX, interactive slicers, KPI cards, trend visualization, topic analysis, event analysis

**EXHIBITS AND
SCHEDULES**

Exhibit A — Executive Overview page (KPI cards, rating & sentiment distribution, top topics)

Exhibit B — Feature Intelligence page (topic mention volume, negative share, average rating by topic)

Exhibit C — Update Impact page (Dark Mode before / event / after comparison)

Exhibit D — Product Action Center page (topic performance summary, insight cards)

REFERENCES

Dataset: Google Play Store user reviews for the Duolingo app (publicly available review export), covering August 19, 2024 – April 9, 2025.

NLP sentiment scoring: VADER (Valence Aware Dictionary and Sentiment Reasoner).